

A Benchmark Framework for Testing, Evaluation, and Comparison of Quadrotor Linear and Nonlinear Controllers

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Abstract—A benchmark framework to test, evaluate, and compare different quadrotor controllers is presented. A detailed nonlinear quadrotor model and a set of six mission scenarios are used to evaluate seven state-of-the-art linear and nonlinear controllers. The quadrotor model is based on the Lagrange formulation and includes aerodynamic and gyroscopic effects, allows for sensor feedback noise to be introduced, and account for first order motor dynamics with input saturation. Simulated mission scenarios include realistic disturbances such as abrupt change of mass, wind gust, and aggressive flight maneuvers. The benchmark framework is the primary contribution of this research; the framework allows for performance comparison of multiple control architectures and implementations, and, the resulting open access testbed is made available to other researchers. Moreover, the same framework may be used to conduct simulated experiments (using ROS/Gazebo, X-Plane, or other software tools), and, with minor modifications, to compare controller performance based on real flights.

Index Terms—Quadrotor, UAV, Modelling, Control, Performance

I. INTRODUCTION

Utilization of quadrotor and multi-rotor Unmanned Aerial Vehicles (UAVs) for civil and public domain applications, in both indoor and outdoor dynamic environments has become increasingly common. Currently published research focuses, among other things, on modeling, navigation and control of such systems [1], as well as on teams of multi-rotor UAVs functioning in unison [2]. The focus in this paper is on developing a framework for realistic comparison of existing controllers. As such, i.) a detailed quadrotor mathematical model is derived following the Lagrange formulation; ii.) the derived model is used as baseline to compare the performance of different commonly-used controllers via simulation of non-aggressive and aggressive maneuvers.

The selected controllers are representative of the most well-known linear and nonlinear controllers cited in the literature [1]. The controllers evaluated are: Proportional Integral Derivative (PID), Linear Quadratic Regulator (LQR), Model Reference Adaptive Control (MRAC), Sliding Mode Control (SMC), H_∞ , inner loop backstepping control (BC), and full backstepping control (BC_{full}).

PID is widely used due to its straightforward implementation and ease of tuning [3], particularly in non-aggressive

maneuvers. A LQR controller aims to steer the dynamics of the quadrotor while minimizing the energy used in the control [4]¹. MRAC exploits the ability to adapt the control gain during flight, and to effectively compensate for external disturbances. The controller tested in this work is formulated according to [5] by deriving the reference model exploiting LQR control as in [6]. SMC is a model based robust controller, which can be used for controlling nonlinear systems. The adopted structure is a centralized version of the one presented in [7]. H_∞ frames the control problem as an optimization problem. Compared to the linear case that involves Riccati equations, the nonlinear case leads to two Hamilton-Jacobi-Bellman-Isaacs partial derivative equations [8]. The formulation evaluated in this work is a nonlinear H_∞ PID controller, introduced in [8] for a robotic manipulator, and tested on a quadrotor in [9]. Backstepping formulations are broadly used for controlling underactuated systems due to their recursive procedure that allows for the controller to overcome coupled and cascaded dynamics [10]. For the benchmark framework, the BC controller is inspired by [11] and two implementations are adopted, one for the inner loop attitude controller (BC), and one for both attitude and position control (BC_{full}).

For the linear controllers (PID, LQR, and MRAC), a feedback linearization approach is used to exactly linearize the nonlinear dynamics of the quadrotor [12]. The simulated missions include noise-free and noise-based scenarios, wind gusts, and mass variation (i.e., a payload release that changes the total quadrotor mass). In contrast to existing studies and surveys [1, 10, 13, 14], this work derives and then uses a single detailed model implemented and tested on all controllers.

The advantage of this benchmark framework is that it allows researchers to test their own derived quadrotor controllers, and compare their performance against the seven controller set. Furthermore, this work allows for testing algorithms on a detailed quadrotor model directly in Matlab & Simulink, making the tuning and testing process faster compared to more complex and resource intensive simulation

¹In this work we implement a linear quadratic trajectory tracking controller but use the LQR acronym in keeping with convention.

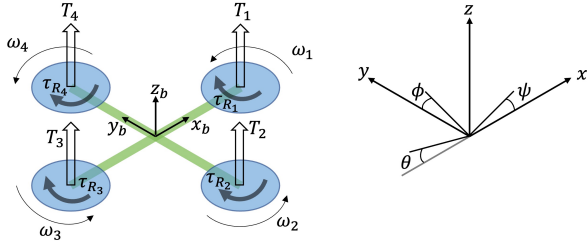


Fig. 1. Quadrotor frames of reference (body frame on the left, inertial frame on the right).

environments. The nonlinear quadrotor model included with the framework is based on the authors' prior work [15], and includes aerodynamic profile and gyroscopic effects. For this work, it is modified to account for further realistic disturbances, such as first-order motor dynamics with input saturation and feedback sensor noise.

The remainder of the paper is organized as follows: Section II presents the detailed quadrotor model. The control approach is described in Section III, while the simulation framework and results are reported in Sections IV and V, respectively. Finally, conclusions and future work are discussed in Section VI.

II. QUADROTOR MODEL

The model is derived following the Lagrange formulation, which has been shown to be the most suitable for model-based control due to its ease in dynamic inversion [12]. Fig. 1, from [15], shows the quadrotor configuration and frame of reference, from which the generalized coordinates representing position, $\xi = [x, y, z]^T$, and Euler angle attitude, $\eta = [\phi, \theta, \psi]^T$, are grouped in the vector $q = [\xi, \eta]^T$.

The equations of motion of the quadrotor model in [15] are presented as

$$B\ddot{q} + C\dot{q} + g + G_a\dot{q} + F\dot{q} = \zeta = K_\zeta\Omega, \quad (1)$$

where:

- $B(q) \in \mathbb{R}^{6 \times 6}$ is the inertia matrix
- $C(q, \dot{q}) \in \mathbb{R}^{6 \times 6}$ is the matrix of centrifugal and Coriolis effects
- $g \in \mathbb{R}^6$ is the gravity vector
- $G_a(q, \dot{q}, \omega) \in \mathbb{R}^{6 \times 6}$ is the gyroscopic matrix
- $F(q, \dot{q}, |\dot{q}|, \omega) \in \mathbb{R}^{6 \times 6}$ is the aerodynamic matrix, accounting for the flapping effect, induced drag, translational drag, profile drag, and parasitic drag where, $A_{flap}, B_{flap}, K_I, K_T, K_P, K_{par}$ are the relative aerodynamic coefficient matrices to be estimated. This aerodynamic model was introduced in [16] and the matrix form is derived in [15] as

$$\begin{bmatrix} D_b \\ \tau_D \end{bmatrix} = F_{flap}(q, \dot{q}, \omega)\dot{q} + F_I(q, \dot{q})\dot{q} + F_T(q, \dot{q})\dot{q} + F_P(q, \dot{q})\dot{q} + F_{par}(q, \dot{q}, |\dot{q}|)\dot{q} = F(q, \dot{q}, |\dot{q}|, \omega)\dot{q}. \quad (2)$$

$F_{flap}, F_I, F_T, F_P, F_{par} \in \mathbb{R}^{6 \times 6}$.

- $\zeta(\omega) \in \mathbb{R}^6$ is the forces and torques vector acting on the quadrotor center of mass. The vector is linearly

dependent on the square of the rotor angular velocity with the following relationship $\zeta = K_\zeta\Omega$, where, ω_i is the angular velocity of the i -th rotor, and K_ζ is found by factoring out $\Omega = [\omega_1^2, \omega_2^2, \omega_3^2, \omega_4^2]^T$ from the thrust and torque model

$$\begin{bmatrix} \zeta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ T \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sum_{i=1}^4 T_i \\ lk(\omega_4^2 - \omega_2^2) \\ lk(\omega_3^2 - \omega_1^2) \\ \tau_{R1} - \tau_{R2} + \tau_{R3} - \tau_{R4} \end{bmatrix}, \quad (3)$$

with $T_i = k\omega_i^2$, $\tau_{Ri} = b\omega_i^2 + I_R\dot{\omega}_i$, k , b , I_R being the lift coefficient, drag constant, and the inertia moment (MOI) of the propeller respectively. Lastly, l represent the arm length of the quadrotor.

A. Motor Dynamics

In a physical quadrotor, the commanded angular velocity is not instantaneously applied to the rotor due to the delay in the electromechanical energy conversion. To account for this, the relation between motor speed the controlled input speed is modelled by the first order dynamic equation:

$$\dot{\omega}_i = \tau_m^{-1}(\omega_{c,i} - \omega_i) \quad (4)$$

where τ_m is the motor time constant and $\omega_{c,i}$ is the control input [17, 18]. In addition, Ω is set as a threshold for the square of the rotor input velocity.

B. Sensor Feedback Noise

Since every real sensor is affected by noise, to increase the fidelity of the simulation environment, Gaussian white noise is injected into the system by adding it to the feedback signal, similar to [19].

C. Mass Drop

To simulate a package drop scenario, the actual mass of the quadrotor starts at $m^* = K_m m$ where m is the nominal mass of the UAV and K_m is the mass increase coefficient. At time t_m the mass immediately changes to the known nominal value.

D. Wind Gust Model

The wind gust model used in the mission scenarios is reported in [20], and described by

$$V_w = \begin{cases} 0 & t < t_i \\ V_{cos} & t_i < t < t_i + T_w \\ 0 & t > t_i + T_w \end{cases} \quad (5)$$

where

- $V_{cos} = (V_{w,max}/2)(1 - \cos(2\pi((t/T_w) - (t_i/T_w))))$
- T_w is the period of the wind gust
- t_i is the initial time of the gust
- $V_{w,max}$ is the wind gust peak
- t is the time expressed in (s)

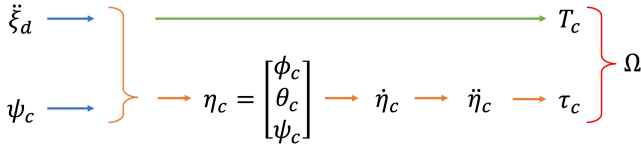


Fig. 2. Hierarchical control structure diagram.

III. CONTROL APPROACH

As background, since a quadrotor is an underactuated system, the matrix K_ζ is not full rank. Thus, a widely followed approach for quadrotor control is to divide the system into two loops:

- Outer control loop which receives as input the reference trajectory $\xi_d = [x_d \ y_d \ z_d]^T$ and outputs the Euler angle reference signal ϕ_c, θ_c ($\psi_c(t) = 0$), and the controlled thrust T_c .
- Inner control loop which receives as input the Euler angle reference signal $\eta_c = [\phi_c \ \theta_c \ \psi_c]^T$ and returns as output the controlled torques τ_c .

This hierarchical structure, shown in Fig. 2, is adopted for all controllers presented in this section except for the full backstepping control (BC_{full}), which exploits the recursive approach to solve the underactuation problem.

The outer loop is designed as a proportional integral derivative (PID) controller similar to [21]. The virtual control vector $U = [U_1, U_2, U_3]^T$ is formulated as

$$U = \ddot{\xi}_d + K_{P_p}(\xi_d - \xi) + K_{d_p}(\dot{\xi}_d - \dot{\xi}) + K_{i_p} \int (\xi_d - \xi) dt. \quad (6)$$

and the resulting reference Euler angles and commanded thrust are given as in [21]

$$\theta_c = \arctan \left(\frac{U_1 \cos \psi + U_2 \sin \psi}{U_3 + g_z} \right), \quad (7)$$

$$\phi_c = \arcsin \left(\frac{U_1 \sin \psi - U_2 \cos \psi}{\sqrt{U_1^2 + U_2^2 + (U_3 + g_z)^2}} \right), \quad (8)$$

$$T_c = m[U_1(\sin \theta \cos \psi \cos \phi + \sin \psi \sin \phi) + U_2(\sin \theta \sin \psi \cos \phi - \cos \psi \sin \phi) + (U_3 + g_z) \cos \theta \cos \phi] \quad (9)$$

As mentioned in Section I, for the linear controllers, feedback linearization is used to exactly linearize the quadrotor nonlinear dynamic [12]. As such, the commanded torque vector is written as

$$\tau_c = J(\eta)v + C_\tau(\eta, \dot{\eta})\dot{\eta}, \quad (10)$$

where $J(\eta)$ is the moment of inertia in the Euler angle coordinates and C_τ is the matrix of centrifugal and Coriolis effects. After dynamic inversion, the closed loop system becomes

$$\ddot{\eta} = v \quad (11)$$

which correspond to the following linear system

$$\dot{x} = Ax + Bv \quad (12)$$

where

$$A = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}, B = \begin{bmatrix} 0_{3 \times 3} \\ I_{3 \times 3} \end{bmatrix}, x = \begin{bmatrix} \eta \\ \dot{\eta} \end{bmatrix} \quad (13)$$

Hence, the virtual control vector v becomes the control action to be designed.

For brevity, the control architectures of the dataset are not reported here but will be part of the documentation of the benchmark framework.

IV. SIMULATION SETUP

The seven controllers are tested in Matlab/Simulink using the model described in Section II. The plant runs at a sample time of $T_s = 1$ ms, the outer loop is set to run at $T_{s,out} = 10$ ms, while the inner loop, which should run at a faster rate than the outer one [19], has a sampling time of $T_{s,in} = 5$ ms. These sampling rates are realistic considering the hardware presented in [19]. The motor threshold for the square of the rotor input velocity $\bar{\Omega}$ is assigned based on the datasheet of “Holybro Brushless Motor 2216-880 KV - CW (520049)”, which would be a feasible motor candidate for the dimension and weight of the proposed quadrotor. The feedback noise variance is set to 0.118 and it introduces an uncertainty of around 1° for the Euler angles feedback signal and 0.02 m for the position feedback signal. The noise is generated by sampling normally distributed random numbers. The mean value is set to 0 and the resulting signal is multiplied by an appropriate coefficient to achieve the desired noise amplitude.

Six mission scenarios are performed for each of the controllers:

- *Base*: Helix tracking with no feedback noise
- *Mass Drop*: Helix tracking with no feedback noise and mass drop at t_m
- *Wind Gust*: Helix tracking with no feedback noise and wind gust at t_i
- *Noise*: Helix tracking with feedback noise
- *Noise Mass Drop*: Helix tracking with feedback noise and mass drop at t_m
- *Noise Wind Gust*: Helix tracking with feedback noise and wind gust at t_i

The helix trajectory is shown in Fig. 3, the diameter is 2 m, and the simulation is performed over a period of 100 s. The quadrotor parameter values are reported in Table I.

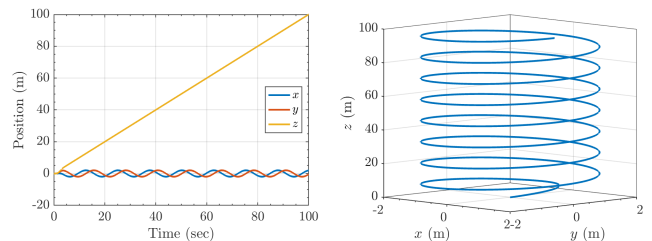


Fig. 3. Helix x, y, z -Trajectory (x, y, z trajectory w.r.t. time on the left, 3D trajectory in space on the right).

TABLE I
SIMULATION PARAMETERS

Param.	Value	Name
m	1.04 kg	Total mass
l	0.2 m	Quadrotor arm length
h	0.168×10^{-1} m	Rotor z-offset from COM
I_x	7.691×10^{-3} kgm ²	Total MOI along x_B
I_y	7.691×10^{-3} kgm ²	Total MOI along y_B
I_z	15.254×10^{-2} kgm ²	Total MOI along z_B
I_{r_x}	1.406×10^{-5} kgm ²	Rotor MOI along x_{rot}
I_{r_y}	1.406×10^{-5} kgm ²	Rotor MOI along y_{rot}
I_{r_z}	2.813×10^{-5} kgm ²	Rotor MOI along z_{rot}
k	1.149×10^{-5}	Thrust coefficient
b	2.690×10^{-8}	Rotor reaction drag coefficient
K_I	1×10^{-2} kgs ⁻¹	Induced drag coefficient
K_{T1}	2.5×10^{-3} kgs ⁻¹	Traslational drag coefficient
K_P	5×10^{-3} kgs ⁻¹	Profile drag coefficient
K_{par}	3×10^{-3} kgs ⁻¹	Parasitic drag coefficient
τ_m	1/20 s	Motor time constant
$\bar{\Omega}$	1102500 rad ² s ⁻²	Rotor squared velocity threshold
V_{wmax}	10 ms ⁻¹	Wind gust peak velocity
K_m	1.5	Mass increase coefficient
t_i	30 s	Wind gust starting time
T_w	20 s	Wind gust period
T_m	30 s	Time of mass drop

The controllers are tuned individually to achieve acceptable performance in the most demanding simulation condition (either the noise mass drop or the noise wind gust scenario) and then the same parameters are use for the other simulations. In practice, the tuning process performed by the authors consists of iteratively changing the respective controller gains to minimize the position and attitude error, avoiding unstable behaviour and control action saturation. All controllers are implemented as presented in literature or are minor modification of such, and no comparison has been done during tuning to avoid biases. Performance is compared through the maximum error (ME) of the generalized coordinates and by considering the root mean square error (RMSE)

$$RMSE = \sqrt{\frac{1}{100} \int_0^{100} \|E\|^2 dt}, \quad (14)$$

where E is either $E_\eta = \eta_c - \eta$ or $E_\xi = \xi_d - \xi$. Since it's important to establish how each controller reacts to disturbances, an analysis of the percentage increase of the RMSE (position) within different scenarios is performed. In particular, the following are considered: Base to Noise, Base to Mass Drop, Base to Wind Gust, Noise to Noise Mass Drop, Noise to Noise Wind Gust.

V. RESULTS

In this section, the results of the benchmark framework are discussed. Table II summarizes results findings. The authors' intent is not to find the "best" control algorithm for quadrotor trajectory tracking but to demonstrate the benefits provided by the benchmark framework in terms of providing interesting insights when the different controllers are compared in a consistent manner under the same scenarios.

Fig. 4 displays the position and attitude error norm of the controllers for the six simulation scenarios. The errors

remain bounded, demonstrating that all controllers are able to achieve stable behaviour. From Fig. 5, 6, 7 it is evident that BC_{full} outperforms all other controllers, suggesting that a complete nonlinear control approach (inner and outer loop) would result in better performance. It is also worth noting that lower RMSE values for orientation do not necessarily guarantee lower RMSE for position, the latter being arguably the most relevant for completing the mission. Between the rest of the controllers, there is no clear best for all scenarios:

- **Base:** the SMC achieves the overall best performance followed closely by the BC and MRAC. The PID is the worst performer and the only one with a maximum position error over 10% of the helix radius.
- **Mass Drop:** all controllers behave similarly in terms of maximum position error but the LQR comes on top in terms of lowest RMSE (position).
- **Wind Gust:** the PID is able to achieve the lowest RMSE and, from Fig. 4, it is evident that it is the least affected by the wind disturbance most likely due to the high gain value on the integral action. The SMC performance, on the other hand, is strongly affected by the wind, which is not expected since robust controllers (like the SMC) are generally more suitable against external disturbances [10]. However, if the disturbance is too severe, the chattering phenomena, associated with the SMC, significantly degrades the controller performance [1], causing it to respond worse than the others.
- **Noise:** the PID is strongly affected by the feedback sensor noise compared to the other controllers, which generally achieve similar performance to the noiseless scenarios. Furthermore, the BC is able to achieve the lowest RMSE (position) value among the rest.
- **Noise Mass Drop:** similarly to the noiseless Mass Drop simulation, the LQR and PID controllers present the lowest and the highest RSME value in terms of position respectively. Furthermore, the SMC performance becomes significantly worse with respect to the noiseless equivalent, for reason explained above.
- **Noise Wind Gust:** different from the noiseless Wind Gust, BC achieves the lowest RMSE (position) while the SMC still presents the largest RMSE values among the group.

From Fig. 8, it's possible to observe and quantify how each controller is affected by the different disturbances. Fig. 8 refers to the percentage performance degradation of the same controller with the addition of disturbances. In fact, it appears that the PID is the one which is most affected by feedback sensor noise, but, on the other hand, it's the least impacted by the mass sudden change and wind gust force. It's important to stress that the PID still performs generally worse than the other controllers in the presence of wind gust and mass drop but its behaviour is more consistent with respect to when these disturbances are not applied. On this note, the SMC behaviour is generally the most susceptible to the added disturbances, again, due to the chattering phenomena. BC and MRAC performance are quite altered by the mass drop scenario even though they still present comparable maximum

TABLE II
RESULTS SUMMARY

Control Law	Advantage	Disadvantage
PID	Simple implementation and good disturbance rejection	Overall worst performance
LQR	Best ease of implementation to performance ratio	Relatively high maximum error
MRAC	Good performance, especially for attitude tracking	Challenging implementation and tuning
SMC	Best performance when no disturbance is applied	Highly affected by external disturbances, filtering is necessary
H_∞	Good performance in presence of disturbances	Relatively challenging implementation
BC	Overall best performance among inner loop controllers	Notably affected by noise and sudden mass change

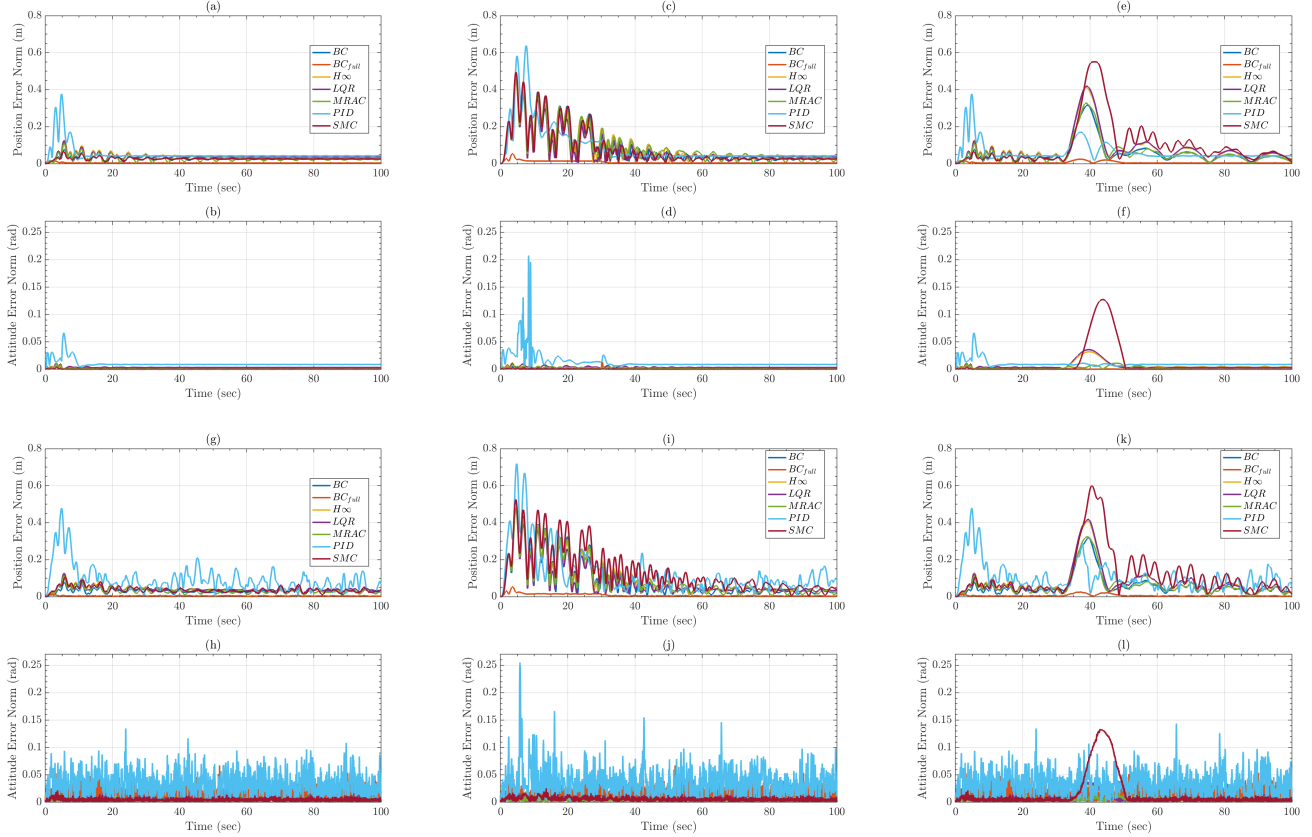


Fig. 4. Position and attitude error norm, horizontal axis represent time expressed in seconds. (a) and (b) are position and attitude error norm respectively for Base, (c) and (d) for Mass Drop, (e) and (f) for Wind Gust, (g) and (h) for Noise, (i) and (j) for Noise Mass Drop, (k) and (l) for Noise Wind Gust.

error to the others, this is not expected for the MRAC since, being an adaptive algorithm, it should respond effectively to parametric uncertainties [10]. However, the sudden change of mass combined with a relatively slow speed of adaptation (required during tuning to maintain stable behaviour) are most probably the reason for this behavior. Furthermore, among all controllers the H_∞ and LQR are only moderately affected by the disturbances. Finally, the BC_{full} controller is evidently the best position tracking performer among the tested algorithms, it is only marginally affected by feedback sensor noise, and it is, in percentage, the second lowest affected by the Mass Drop and Wind Gust.

VI. CONCLUSIONS

A comparison framework for testing and evaluating quadrotor controller performance has been presented. A set of

seven controllers has been implemented and tested using the same detailed quadrotor model derived using the Lagrangian formulation. The performed simulations included realistic mission disturbances such as sensor noise, mass drop, and wind gust. The performance has been evaluated through RMSE (position and orientation) and maximum error when tracking a helix trajectory. From the results, it has been observed that a fully nonlinear controller performs better than the more classic hierarchical structure composed of a PID outer loop and a designed controller for the inner loop. However, among the tested controllers with this cascade structure, the BC was generally the one with best performance, followed by the MRAC. It is worth noting that, even if the PID generally performs the worst and returns the highest maximum position error, it is the most “consistent” against disturbances such as mass variation and wind gust. Consid-

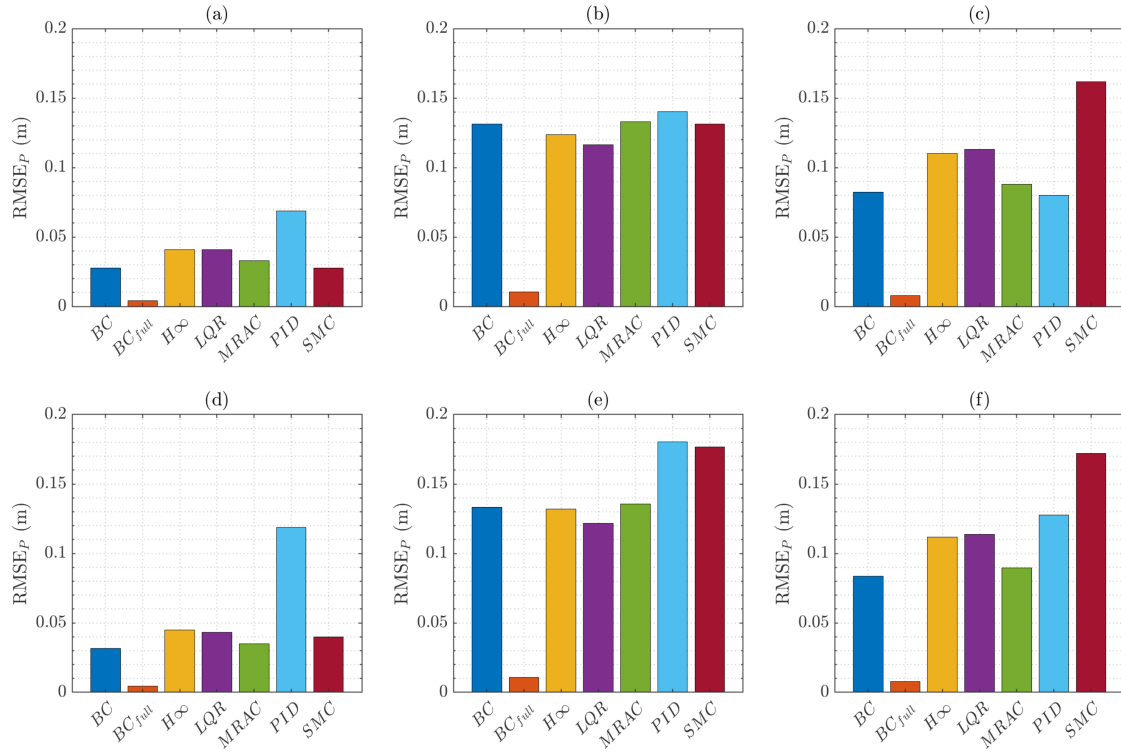


Fig. 5. $RMSE_P$ comparison between controllers (Horizontal axis). Simulation scenario: (a) Base, (b) Mass Drop, (c) Wind Gust, (d) Noise, (e) Noise Mass Drop, (f) Noise Wind Gust.

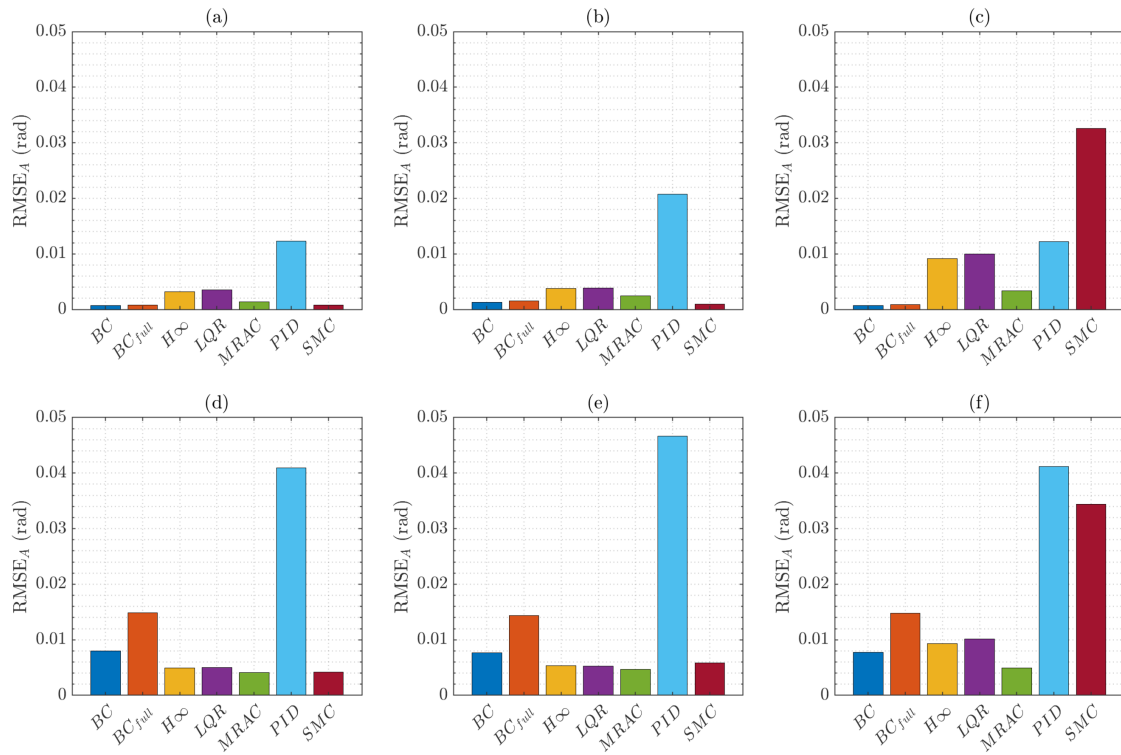


Fig. 6. $RMSE_A$ comparison between controllers (Horizontal axis). Simulation scenario: (a) Base, (b) Mass Drop, (c) Wind Gust, (d) Noise, (e) Noise Mass Drop, (f) Noise Wind Gust.

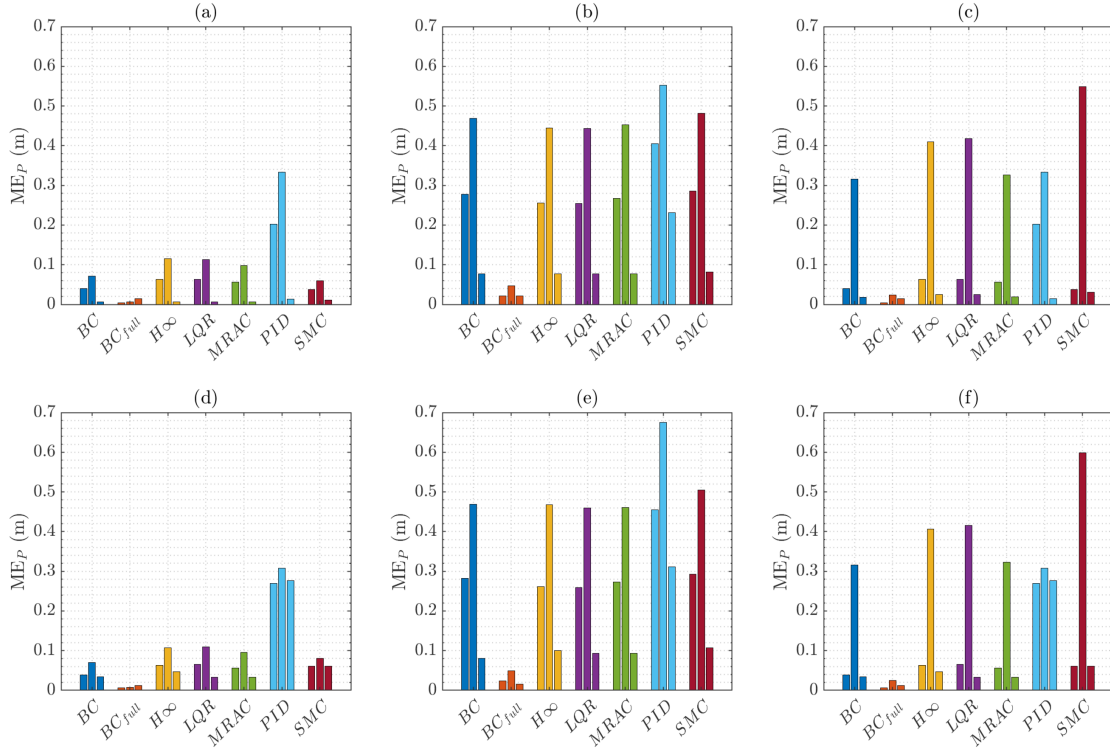


Fig. 7. ME_P comparison between controllers (Horizontal axis). Each controller has 3 bars representing x, y, z coordinates respectively. Simulation scenario: (a) Base, (b) Mass Drop, (c) Wind Gust, (d) Noise, (e) Noise Mass Drop, (f) Noise Wind Gust.

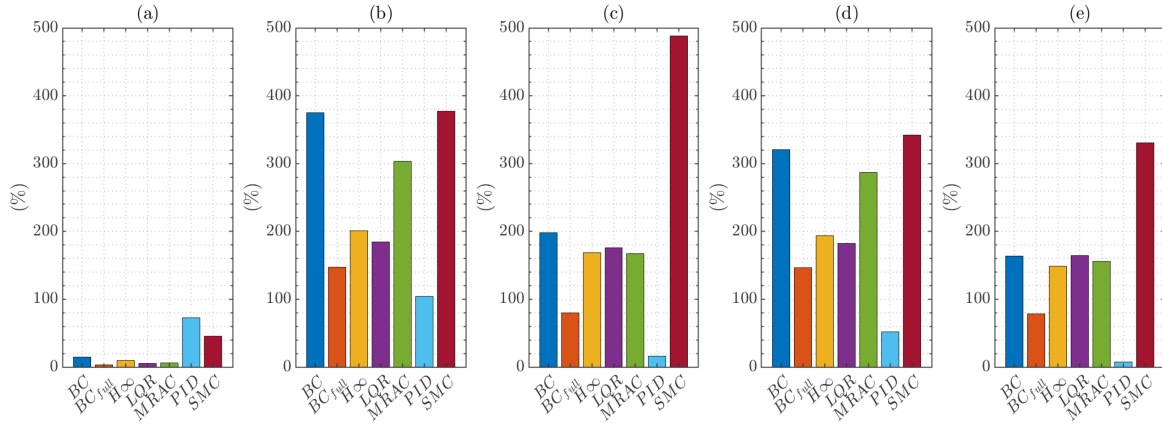


Fig. 8. $RMSE_P$ % increase within different simulation scenarios: (a) Base to Noise, (b) Base to Mass Drop, (c) Base to Wind Gust, (d) Noise to Noise Mass Drop, (e) Noise to Noise Wind Gust.

ering the difficulty of realization and tuning, the nonlinear controllers, and the MRAC particularly, generally take more time and trials to implement and find the adequate gains and parameter values. In this respect, the LQR stands out, among the rest, for ease of implementation to performance ratio. With that said, all controllers are able to achieve relatively satisfactory performance and remain stable throughout the simulation (longer periods of time than 100 s have been tested to confirm this) and, considering that all simulation times are faster than real time, it is auspicious for future real time simulated experiments.

This framework allows for a fair comparison of control

strategies under different simulation scenarios, highlighting that, in practice, theoretical performance may not be met due to presence of noise, external disturbances, model mismatches, and realistic sample time. Moreover, it provides an extensive database to be used as a benchmark to evaluate quadrotor controllers.

In ongoing and future work, a wider family of representative controllers will be tested, including hybrid configurations of the ones already tested, as well as different hierarchical configurations of inner and outer loop. It is interesting to investigate the performance of feedback linearization accounting for the full nonlinear dynamics (i.e considering

gyroscopic and aerodynamic effects) compared to the one performed in this work and the equivalent pure linear form of the controllers. Also, additional performance indexes such as steady state error, overshoot, and control action maximum and integral values could be used for a more complete performance analysis. It is worth noting that the tuning process may be standardized using nonlinear controller tuning strategies such as Iterative Feedback Tuning (IFT). In addition, it would be useful to add new mission scenarios for testing other types of disturbances and perform a comparison of the same mission scenario with respect to a spectrum of values of the disturbance influence. Furthermore, the model accuracy may be further improved by adding realistic models of gyroscope, accelerometer, compass, and GPS and considering ground, wall, and ceiling aerodynamic effects. Finally, the authors plan on presenting an equivalent benchmark framework in ROS to further analyze real time performance.

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